SAMS — Sprint 1 Test Results & Defects

TASK-004 | Execution summary for TC-001 to TC-045

Project: Student Attendance Management System (SAMS)

Environment: Local FastAPI backend, static frontend, and SQLite test database.

1. Execution summary

Core workflows, validation, authorization, attendance correction, reporting, password hashing and session expiry were exercised locally. Cases needing deployment or mobile evidence are explicitly qualified.

Status	Summary
PASS	Expected behavior observed for most functional, validation, authorization and session checks.
PASS WITH OBSERVATION / LIMITED SCOPE	TC-008, TC-033, TC-034, TC-038 and TC-039 include transient-error, warning or smoke-test limitations.
PARTIALLY VERIFIED	TC-037: desktop UI verified, mobile responsiveness outstanding. TC-041: hashing verified, comprehensive log redaction not proven.
PENDING DEPLOYMENT VERIFICATION	TC-040 HTTPS/TLS.
PASS FOR TESTED CHECKS	TC-043 invalid-input test and inspected parameterized search query; not a full audit of every endpoint.

2. Detailed test results

ID	Test case	Observed result	Status / note
TC-001	Admin valid login	HTTP 200; token and Admin role	PASS
TC-002	Admin invalid password	HTTP 401	PASS
TC-003	Create class	HTTP 200; Test Class A, ID 1	PASS
TC-004	Create student	HTTP 200; Test Student One, ID 1	PASS
TC-005	List students	HTTP 200; active student returned	PASS
TC-006	Invalid student email	HTTP 400	PASS
TC-007	Duplicate student email	HTTP 409	PASS
TC-008	Create subject	Initial HTTP 500 SQLite lock; retry HTTP 200, ID 1	PASS WITH OBSERVATION
TC-009	Create faculty	HTTP 200; faculty ID 1	PASS
TC-010	Create assignment	HTTP 200; assignment ID 1	PASS
TC-011	Create attendance session	HTTP 200; session ID 1	PASS
TC-012	Mark attendance Present	HTTP 200; one record saved	PASS
TC-013	Duplicate attendance prevention	HTTP 409 Duplicate attendance	PASS
TC-014	Attendance history	HTTP 200; history returned	PASS
TC-015	Attendance percentage	HTTP 200; 100%, 1/1 before correction	PASS
TC-016	Low-attendance list	HTTP 200; empty before correction	PASS
TC-017	Attendance report	HTTP 200	PASS
TC-018	CSV export	HTTP 200	PASS
TC-019	Filter by Present status	HTTP 200 before correction	PASS
TC-020	Correct Present to Absent	HTTP 200	PASS
TC-021	Audit log for correction	HTTP 200; correct_Present_to_Absent visible	PASS
TC-022	Invalid attendance status	HTTP 400	PASS
TC-023	Unknown session ID	HTTP 404 Session not found	PASS
TC-024	Student not enrolled in class	HTTP 400	PASS
TC-025	Attendance rules	HTTP 200; threshold 75%, correction window 48h	PASS
TC-026	Date filter	HTTP 200	PASS
TC-027	Student access to admin audit	HTTP 403	PASS
TC-028	Logout invalidates token	Logout HTTP 200; reused token HTTP 401	PASS
TC-029	Unauthenticated student list	HTTP 401	PASS
TC-030	Faculty login	HTTP 200; role Faculty	PASS
TC-031	Faculty attendance summary	HTTP 200	PASS
TC-032	Faculty attendance report	HTTP 200	PASS
TC-033	Response-time smoke test	/openapi.json HTTP 200 in 0.001702 seconds (single local request)	PASS — LIMITED SCOPE
TC-034	Availability/recovery smoke test	HTTP 200 after backend restart	PASS — LIMITED SCOPE
TC-035	Password storage	Salted PBKDF2-HMAC-SHA256, 200,000 iterations; salt$hash	PASS
TC-036	Attendance data integrity	Duplicate mark returned HTTP 409	PASS
TC-037	UI login/dashboard	Valid login opened dashboard; 0%, 0 of 1 attended; history shows Absent for 2026-10-26, period 1, Test Subject	PARTIALLY VERIFIED — mobile not separately tested
TC-038	Maintainability and automated tests	pytest: 2 passed, 18 warnings; main logic concentrated in backend/main.py	PASS WITH OBSERVATION
TC-039	Concurrent requests smoke test	Five simultaneous /openapi.json requests all HTTP 200	PASS — LIMITED SCOPE
TC-040	HTTPS/TLS	HTTP 200; HTTPS request HTTP 000 on	PENDING DEPLOYMENT

		local dev server	VERIFICATION
TC-041	Password hashing and log redaction	Hashing verified; no print/logging/logger/basicConfig matches found in backend/main.py	PARTIALLY VERIFIED — log check limited to one file
TC-042	Role-based authorization	Student token to admin-only /api/audit returned HTTP 403	PASS
TC-043	Input validation and injection	Invalid student data HTTP 400; injection-style search returned [] / HTTP 200; inspected search query uses placeholders and args	PASS FOR TESTED CHECKS
TC-044	Protected audit logs	Student token to /api/audit returned HTTP 403	PASS
TC-045	Session timeout	With SAMS_SESSION_TIMEOUT=5, reused token after 7 seconds returned HTTP 401 Session expired	PASS

3. Defect and observation register

Ref / TC	Observation	Evidence	Disposition / next action
OBS-001 / TC-008	Transient SQLite lock during subject creation	First request HTTP 500; retry HTTP 200, subject ID 1.	Investigate if reproducible.
OBS-002 / TC-037	Frontend API URL integration / responsive UI	Initial frontend request hit static server (HTTP 501); after switching to fix/api-url, valid student login opened dashboard on Mac.	Desktop login resolved; separately verify mobile. Phone localhost points to phone, not Mac.
OBS-003 / TC-038	Automated test warnings	pytest: 2 passed, 18 warnings, including deprecation warnings.	Review deprecated FastAPI on_event and datetime.utcnow usage.
OBS-004 / TC-040	HTTPS/TLS unavailable on local dev server	HTTP endpoint returned 200; HTTPS request returned HTTP 000.	Pending deployment verification; verify actual deployment.
OBS-005 / TC-041	Log redaction evidence incomplete	No print/logging/logger/basicConfig matches found in backend/main.py.	Inspect all modules and server/framework logs.
OBS-006 / TC-033, TC-039	Performance/concurrency evidence limited	Single OpenAPI request 0.001702 seconds; five concurrent OpenAPI requests returned 200.	Smoke checks only; formal load testing if required.

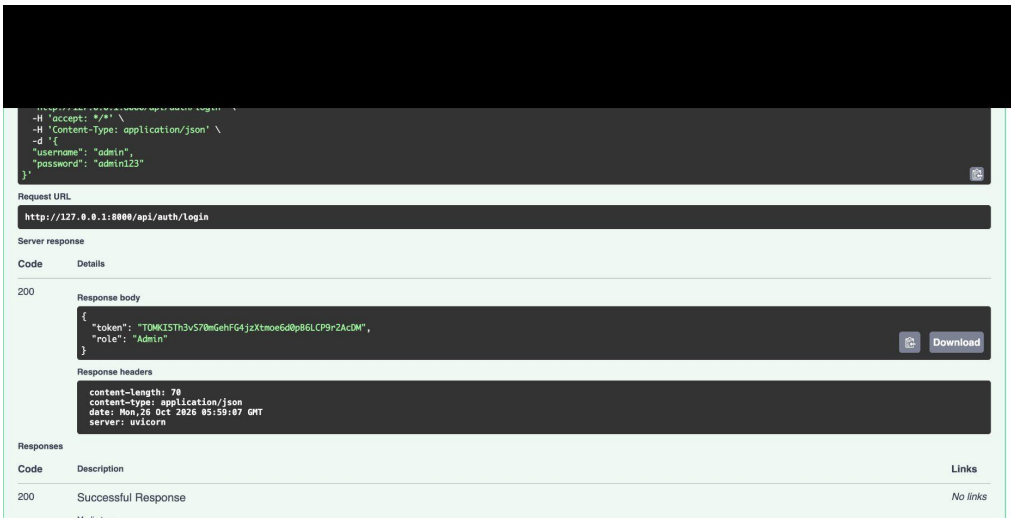
4. Evidence register

Evidence ID	Item	Recorded details	Interpretation
EV-001 to EV-040	40 screenshots supplied earlier in the conversation	Embedded in Section 5 below as Evidence_01.jpg through Evidence_40.jpg.	Retained as user-provided evidence. Specific test mapping should be cross-checked against original execution sequence.
EV-041	Latest student dashboard screenshot	Screenshot 2026-10-27 at 21.12.58.png	Successful student dashboard after login; 0% attendance; 0 of 1 classes attended; history row shows Absent.

5. Screenshot evidence (all supplied evidence images)

The following pages embed the 40 screenshots supplied earlier, followed by the latest dashboard screenshot. Evidence images are reproduced as received; labels identify the source image rather than asserting an unverified TC mapping.

Evidence_01.jpg



Evidence_02.jpg



```
  "email": "student1@example.com",
  "password": "StudentTest@123",
  "class_id": 1
}'
```

Request URL

```
http://127.0.0.1:8000/api/students
```

Server response

Code	Details
401 *Undocumented*	Error: Unauthorized

Response body

```
{
  "detail": "Not authenticated"
}
```

Download

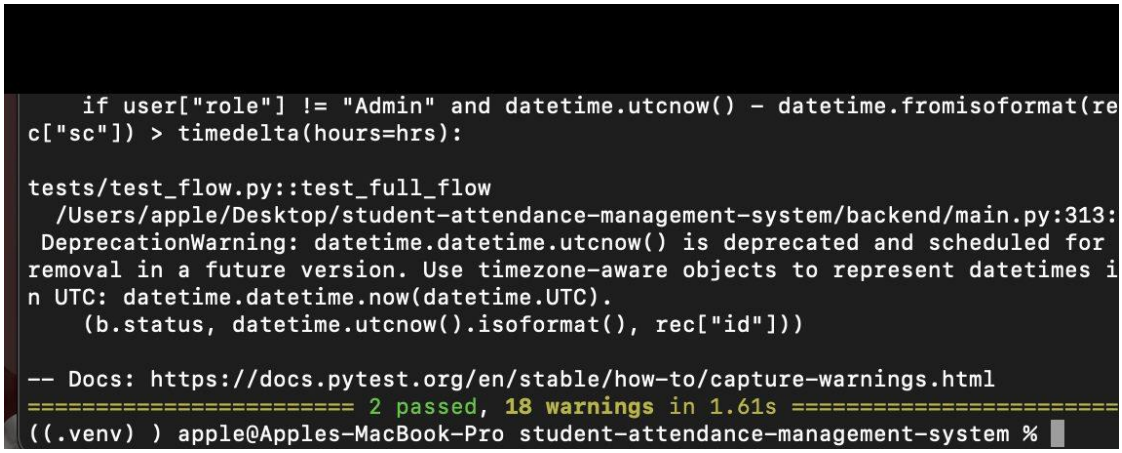
Response headers

```
content-length: 30
content-type: application/json
date: Mon,26 Oct 2026 06:10:36 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Evidence_03.jpg



Evidence_04.jpg



Evidence_05.jpg



Evidence_06.jpg



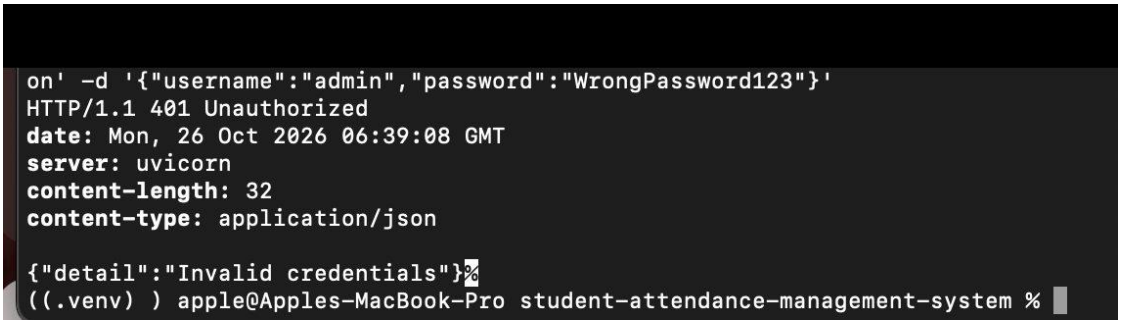
Evidence_07.jpg



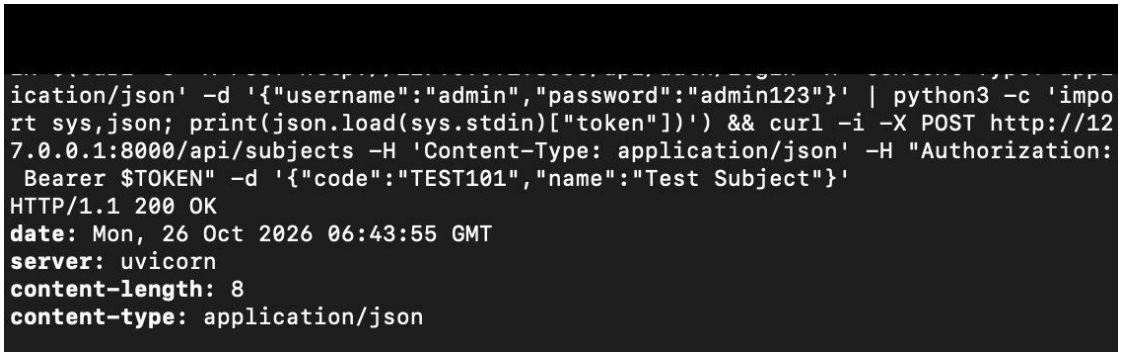
Evidence_08.jpg



Evidence_09.jpg

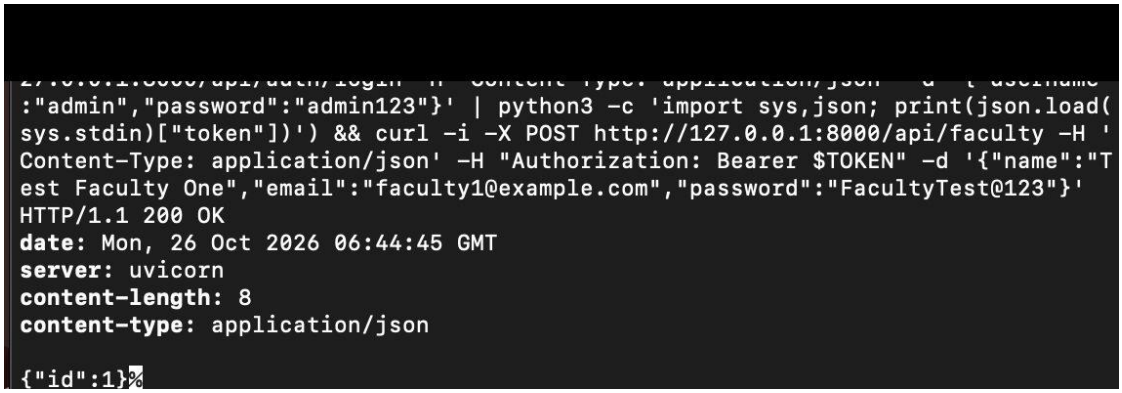


Evidence_10.jpg

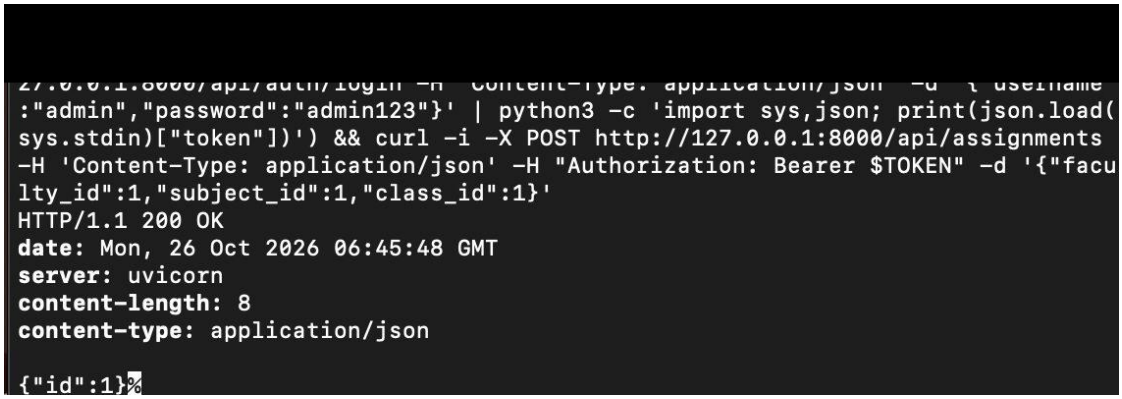


```
ication/json' -d '{"username":"admin","password":"admin123"}' | python3 -c 'impo
rt sys,json; print(json.load(sys.stdin)["token"])') && curl -i -X POST http://12
7.0.0.1:8000/api/subjects -H 'Content-Type: application/json' -H "Authorization:
 Bearer $TOKEN" -d '{"code":"TEST101","name":"Test Subject"}'
HTTP/1.1 200 OK
date: Mon, 26 Oct 2026 06:43:55 GMT
server: uvicorn
content-length: 8
content-type: application/json
```

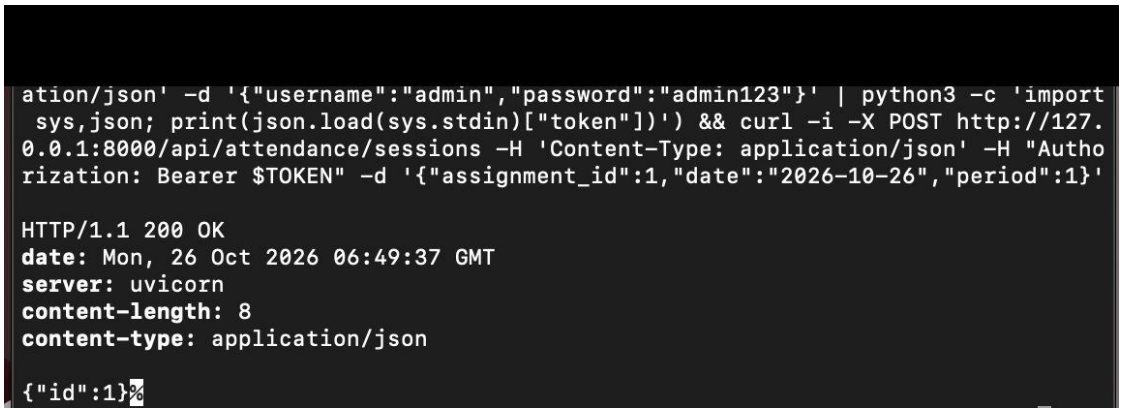
Evidence_11.jpg



Evidence_12.jpg



Evidence_13.jpg

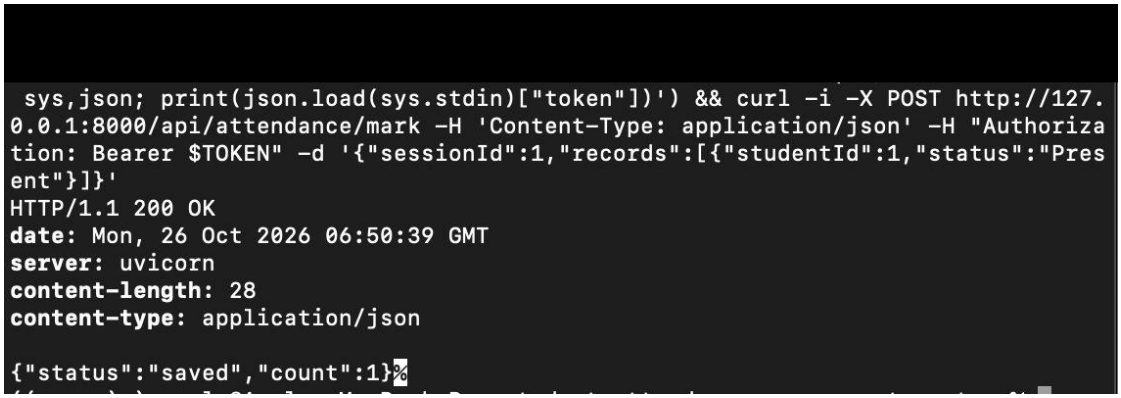


```
ation/json' -d '{"username":"admin","password":"admin123"}' | python3 -c 'import
 sys,json; print(json.load(sys.stdin)["token"])') && curl -i -X POST http://127.
0.0.1:8000/api/attendance/sessions -H 'Content-Type: application/json' -H "Autho
rization: Bearer $TOKEN" -d '{"assignment_id":1,"date":"2026-10-26","period":1}'

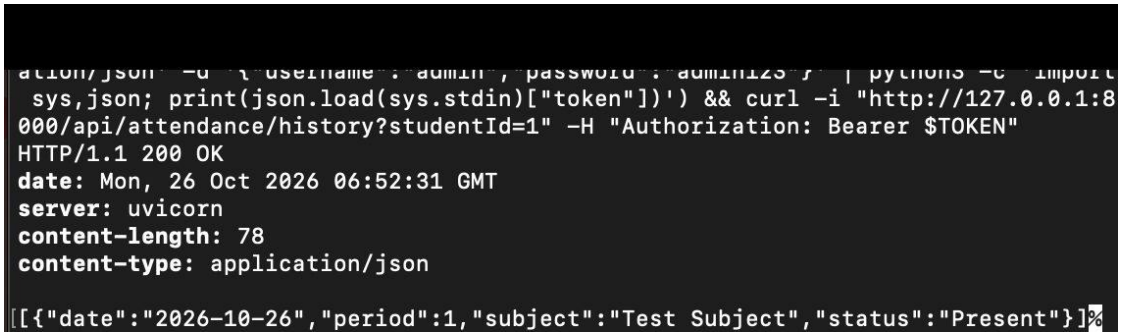
HTTP/1.1 200 OK
date: Mon, 26 Oct 2026 06:49:37 GMT
server: uvicorn
content-length: 8
content-type: application/json

{"id":1}%
```

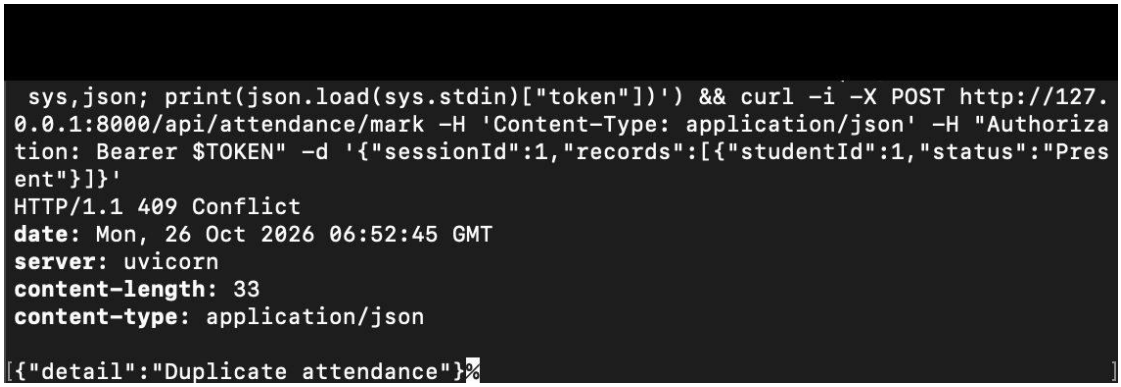
Evidence_14.jpg



Evidence_15.jpg



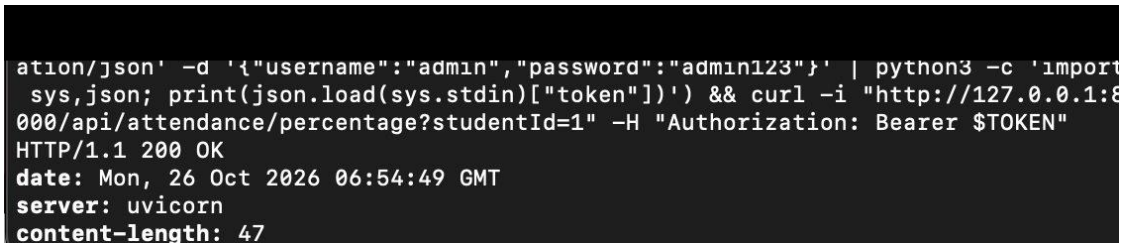
Evidence_16.jpg



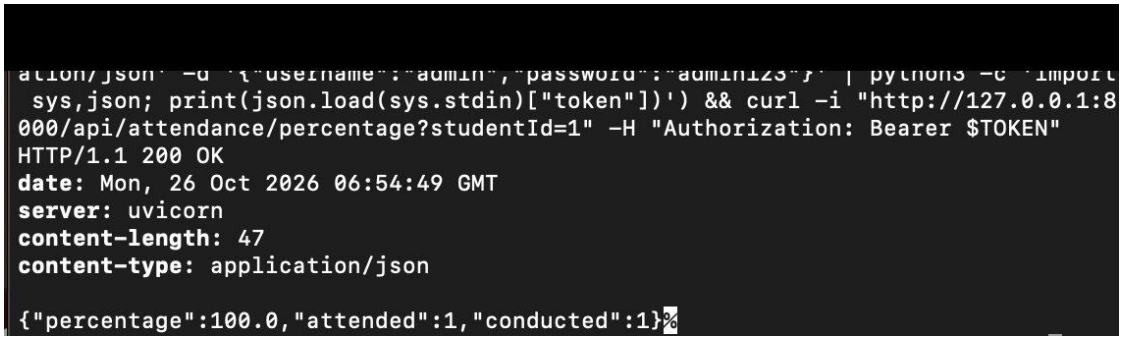
```
 sys,json; print(json.load(sys.stdin)["token"])') && curl -i -X POST http://127.
0.0.1:8000/api/attendance/mark -H 'Content-Type: application/json' -H "Authoriza
tion: Bearer $TOKEN" -d '{"sessionId":1,"records":[{"studentId":1,"status":"Pres
ent"}]}'
HTTP/1.1 409 Conflict
date: Mon, 26 Oct 2026 06:52:45 GMT
server: uvicorn
content-length: 33
content-type: application/json

{"detail":"Duplicate attendance"}%
```

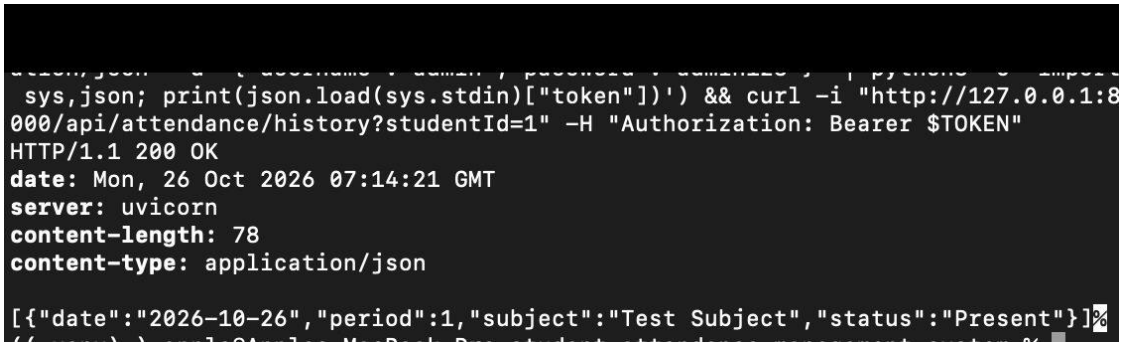
Evidence_17.jpg



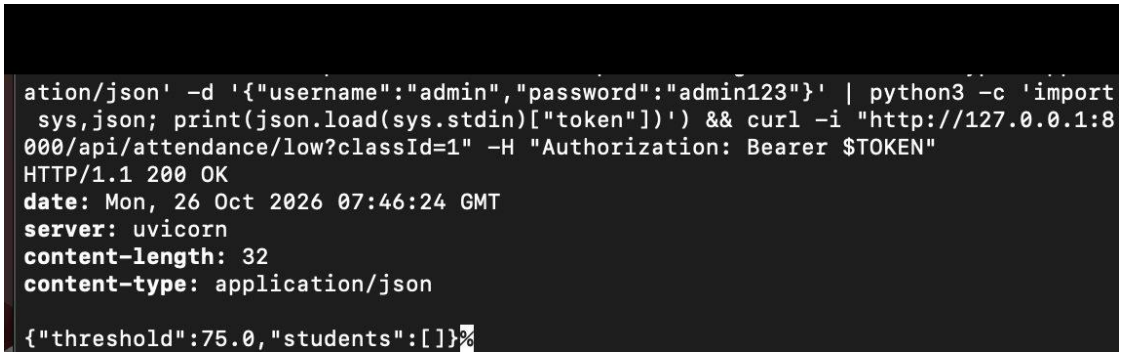
Evidence_18.jpg



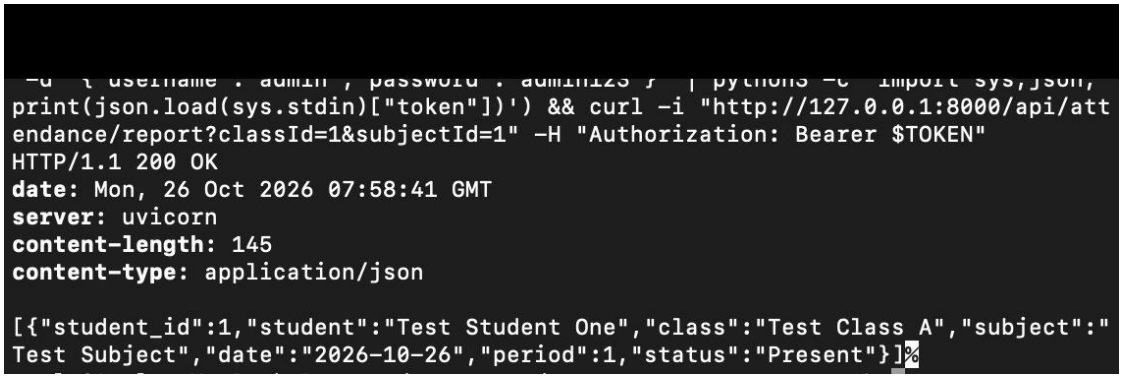
Evidence_19.jpg



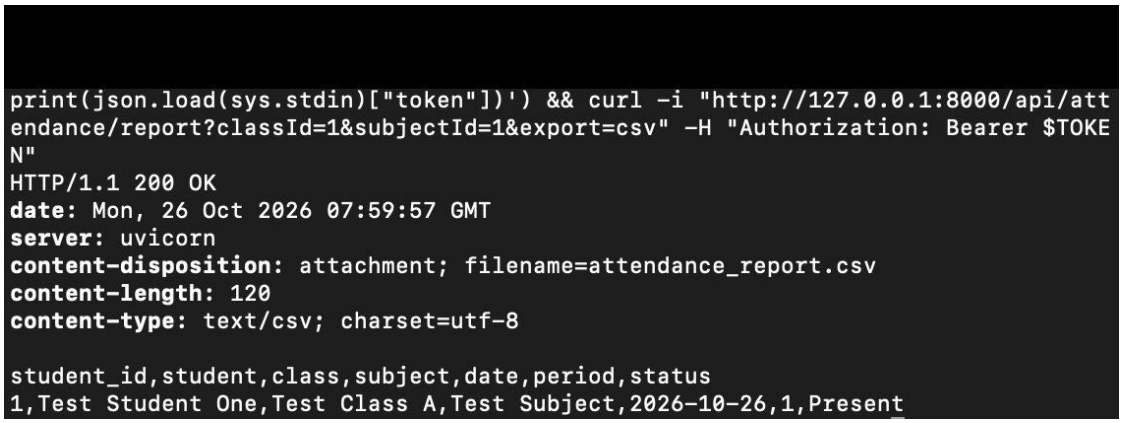
Evidence_20.jpg



Evidence_21.jpg



Evidence_22.jpg



```
print(json.load(sys.stdin)["token"])') && curl -i "http://127.0.0.1:8000/api/att
endance/report?classId=1&subjectId=1&export=csv" -H "Authorization: Bearer $TOKE
N"
HTTP/1.1 200 OK
date: Mon, 26 Oct 2026 07:59:57 GMT
server: uvicorn
content-disposition: attachment; filename=attendance_report.csv
content-length: 120
content-type: text/csv; charset=utf-8

student_id,student,class,subject,date,period,status
1,Test Student One,Test Class A,Test Subject,2026-10-26,1,Present
```

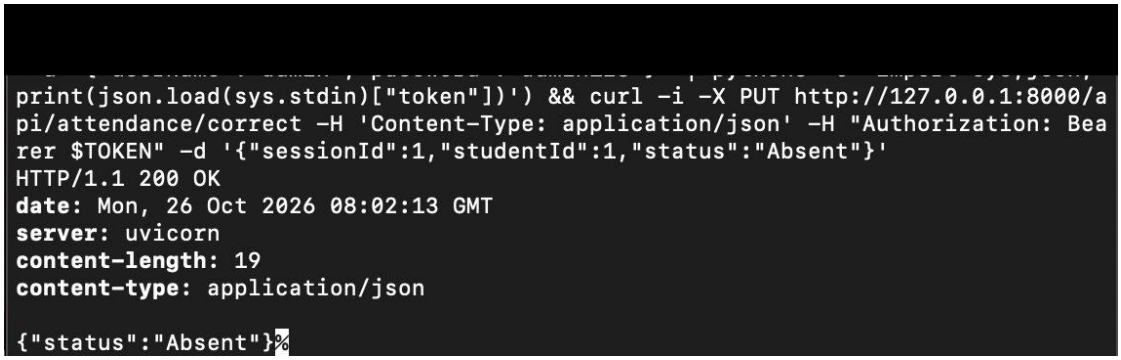
Evidence_23.jpg



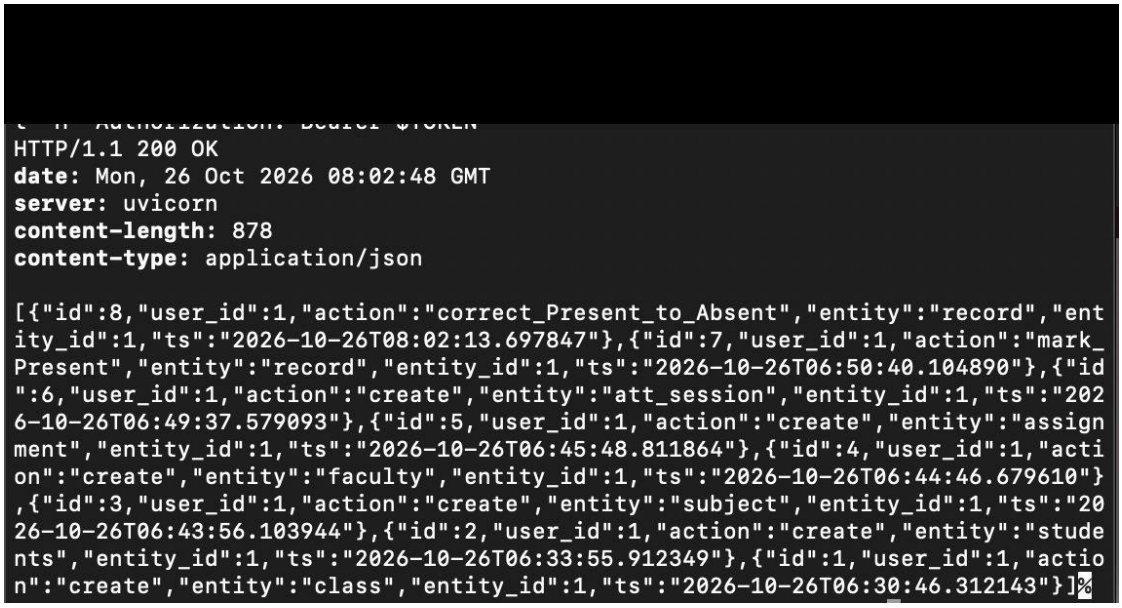
```
print(json.load(sys.stdin)["token"])') && curl -i "http://127.0.0.1:8000/api/att
endance/report?classId=1&subjectId=1&status=Present" -H "Authorization: Bearer $
TOKEN"
HTTP/1.1 200 OK
date: Mon, 26 Oct 2026 08:00:46 GMT
server: uvicorn
content-length: 145
content-type: application/json

[{"student_id":1,"student":"Test Student One","class":"Test Class A","subject":"
Test Subject","date":"2026-10-26","period":1,"status":"Present"}]%
```

Evidence_24.jpg



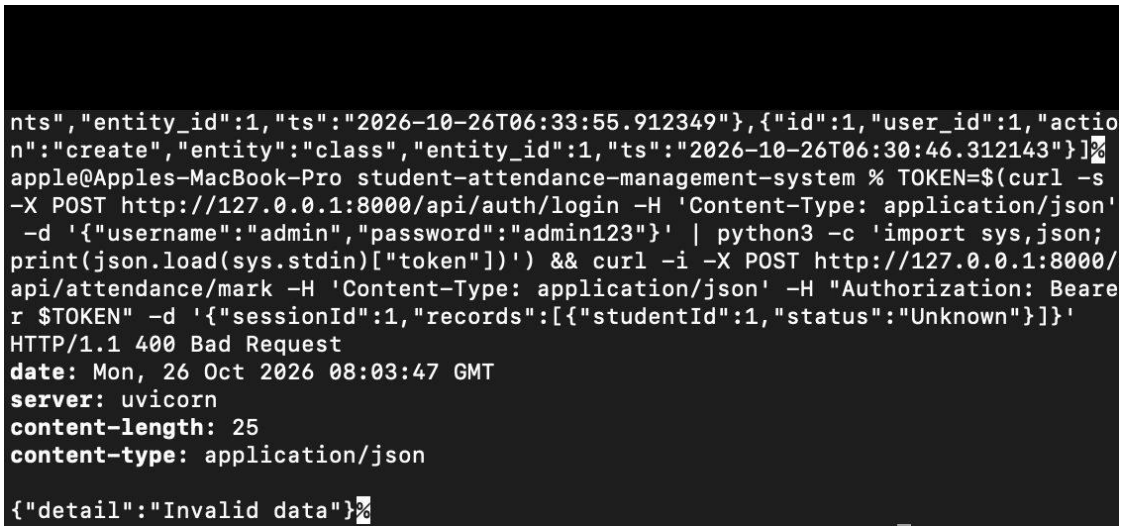
Evidence_25.jpg



```
HTTP/1.1 200 OK
date: Mon, 26 Oct 2026 08:02:48 GMT
server: uvicorn
content-length: 878
content-type: application/json

[{"id":8,"user_id":1,"action":"correct_Present_to_Absent","entity":"record","ent
ity_id":1,"ts":"2026-10-26T08:02:13.697847"},{"id":7,"user_id":1,"action":"mark_
Present","entity":"record","entity_id":1,"ts":"2026-10-26T06:50:40.104890"},{"id
":6,"user_id":1,"action":"create","entity":"att_session","entity_id":1,"ts":"202
6-10-26T06:49:37.579093"},{"id":5,"user_id":1,"action":"create","entity":"assign
ment","entity_id":1,"ts":"2026-10-26T06:45:48.811864"},{"id":4,"user_id":1,"acti
on":"create","entity":"faculty","entity_id":1,"ts":"2026-10-26T06:44:46.679610"}
,{"id":3,"user_id":1,"action":"create","entity":"subject","entity_id":1,"ts":"20
26-10-26T06:43:56.103944"},{"id":2,"user_id":1,"action":"create","entity":"stude
nts","entity_id":1,"ts":"2026-10-26T06:33:55.912349"},{"id":1,"user_id":1,"actio
n":"create","entity":"class","entity_id":1,"ts":"2026-10-26T06:30:46.312143"}]%
```

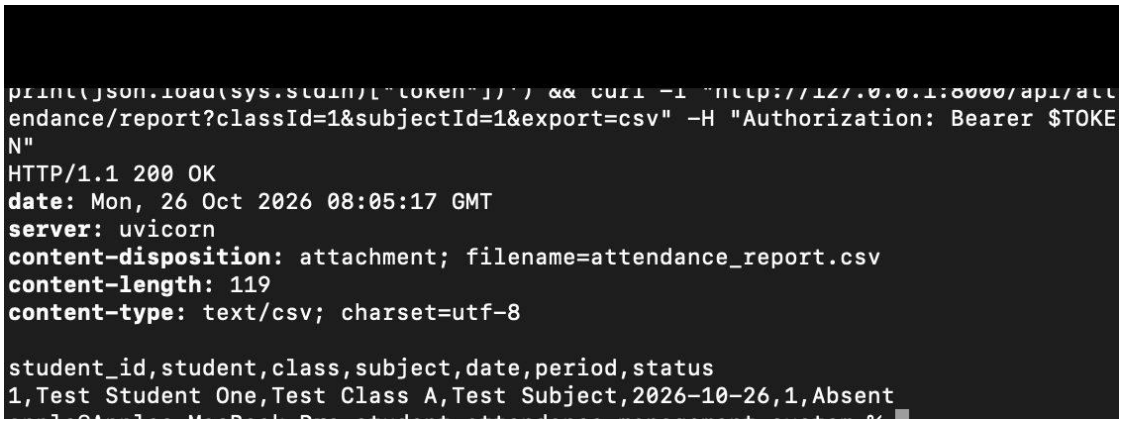
Evidence_26.jpg



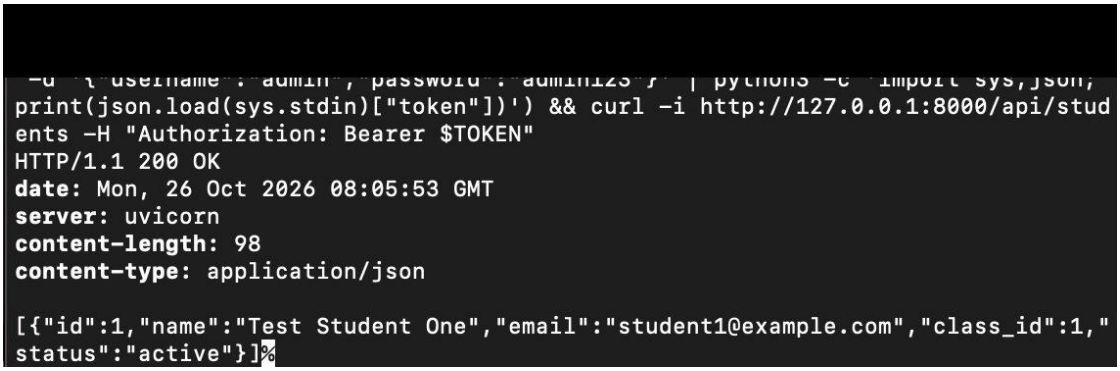
```
nts","entity_id":1,"ts":"2026-10-26T06:33:55.912349"},{"id":1,"user_id":1,"actio
n":"create","entity":"class","entity_id":1,"ts":"2026-10-26T06:30:46.312143"}]%
apple@Apples-MacBook-Pro student-attendance-management-system % TOKEN=$(curl -s
-X POST http://127.0.0.1:8000/api/auth/login -H 'Content-Type: application/json'
 -d '{"username":"admin","password":"admin123"}' | python3 -c 'import sys,json;
print(json.load(sys.stdin)["token"])') && curl -i -X POST http://127.0.0.1:8000/
api/attendance/mark -H 'Content-Type: application/json' -H "Authorization: Beare
r $TOKEN" -d '{"sessionId":1,"records":[{"studentId":1,"status":"Unknown"}]}'
HTTP/1.1 400 Bad Request
date: Mon, 26 Oct 2026 08:03:47 GMT
server: uvicorn
content-length: 25
content-type: application/json

{"detail":"Invalid data"}%
```

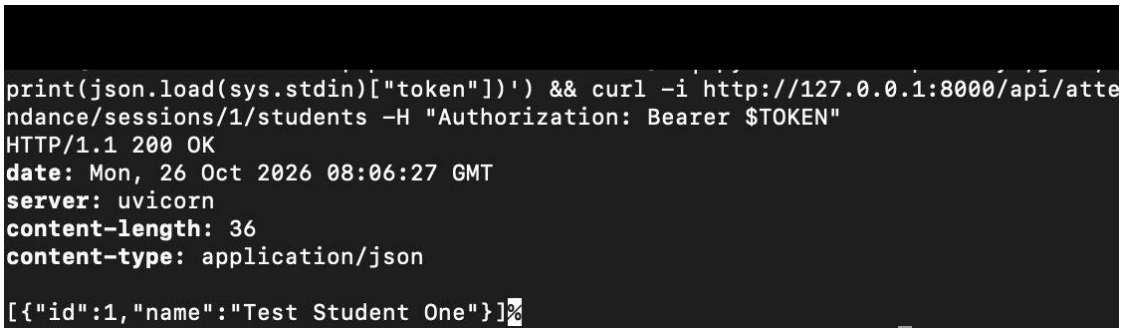
Evidence_27.jpg



Evidence_28.jpg



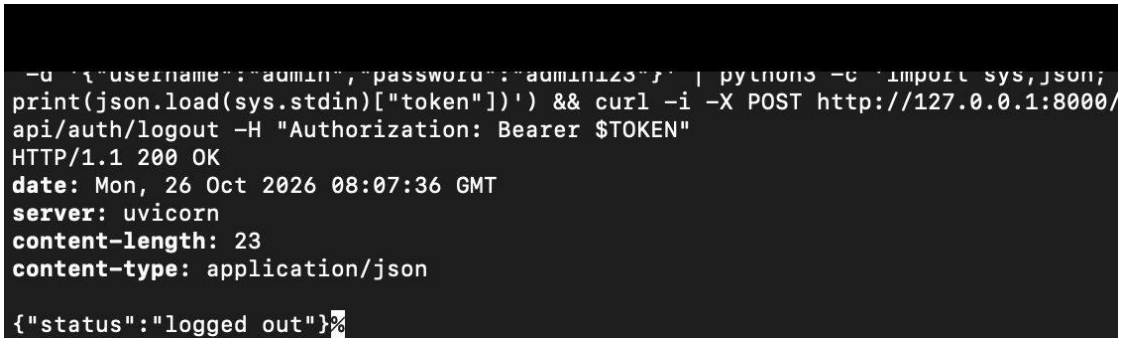
Evidence_29.jpg



Evidence_30.jpg



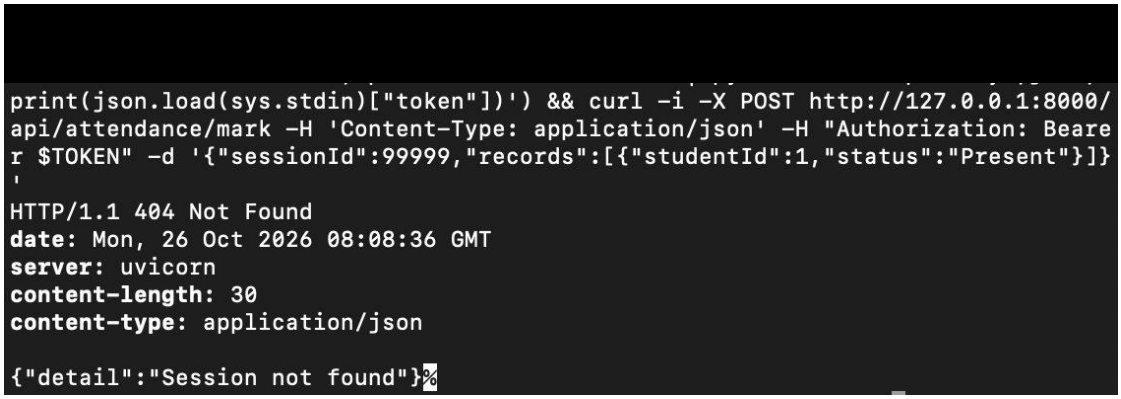
Evidence_31.jpg



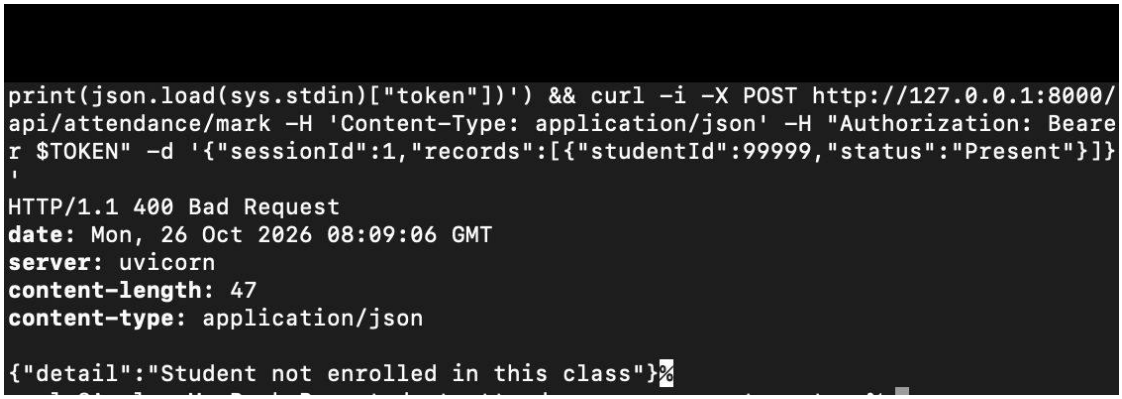
Evidence_32.jpg



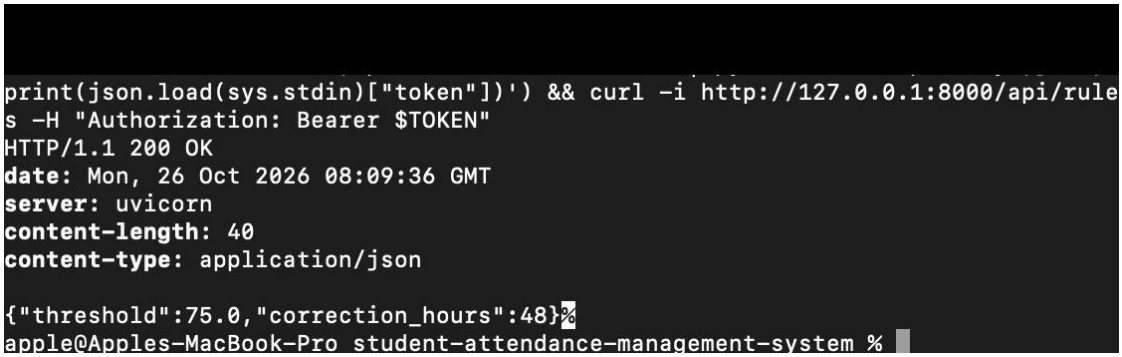
Evidence_33.jpg



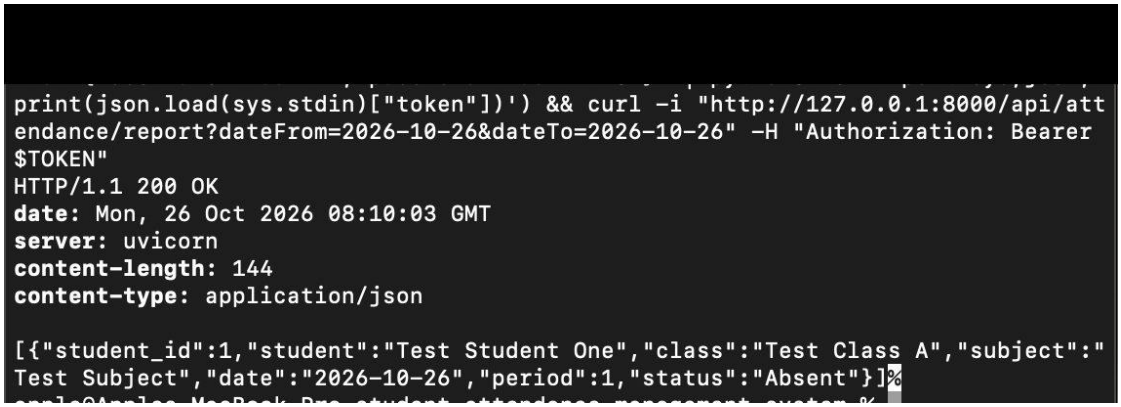
Evidence_34.jpg



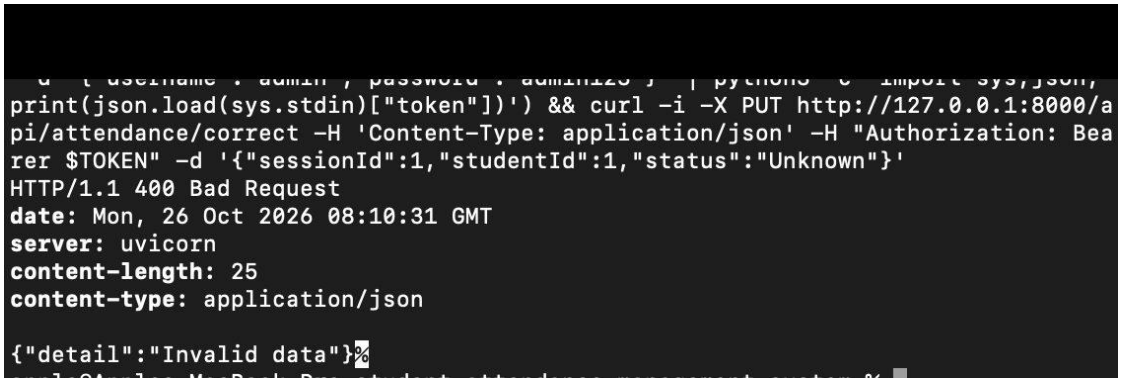
Evidence_35.jpg



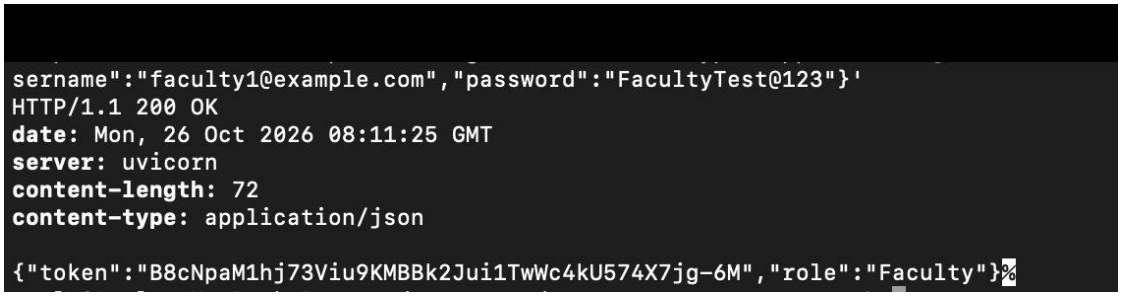
Evidence_36.jpg



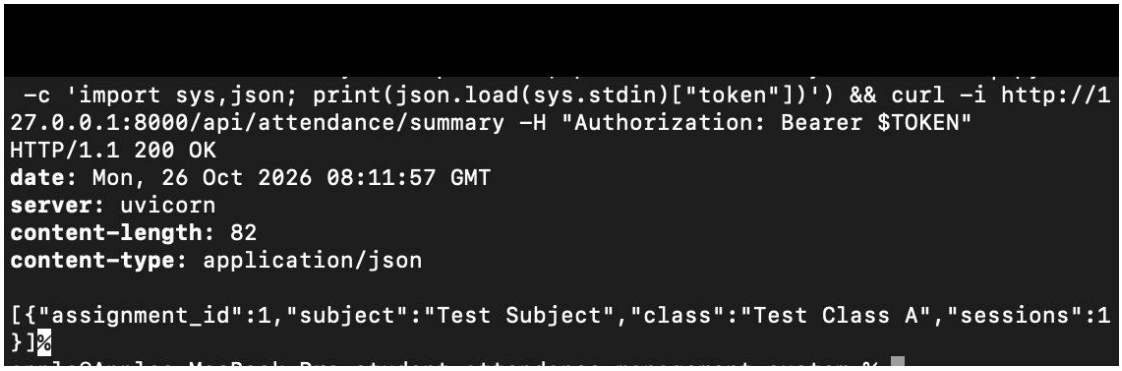
Evidence_37.jpg



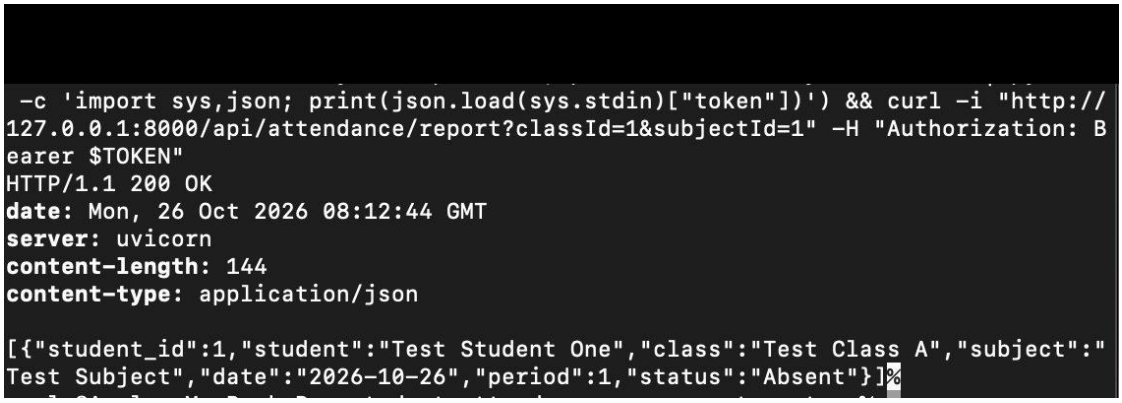
Evidence_38.jpg



Evidence_39.jpg



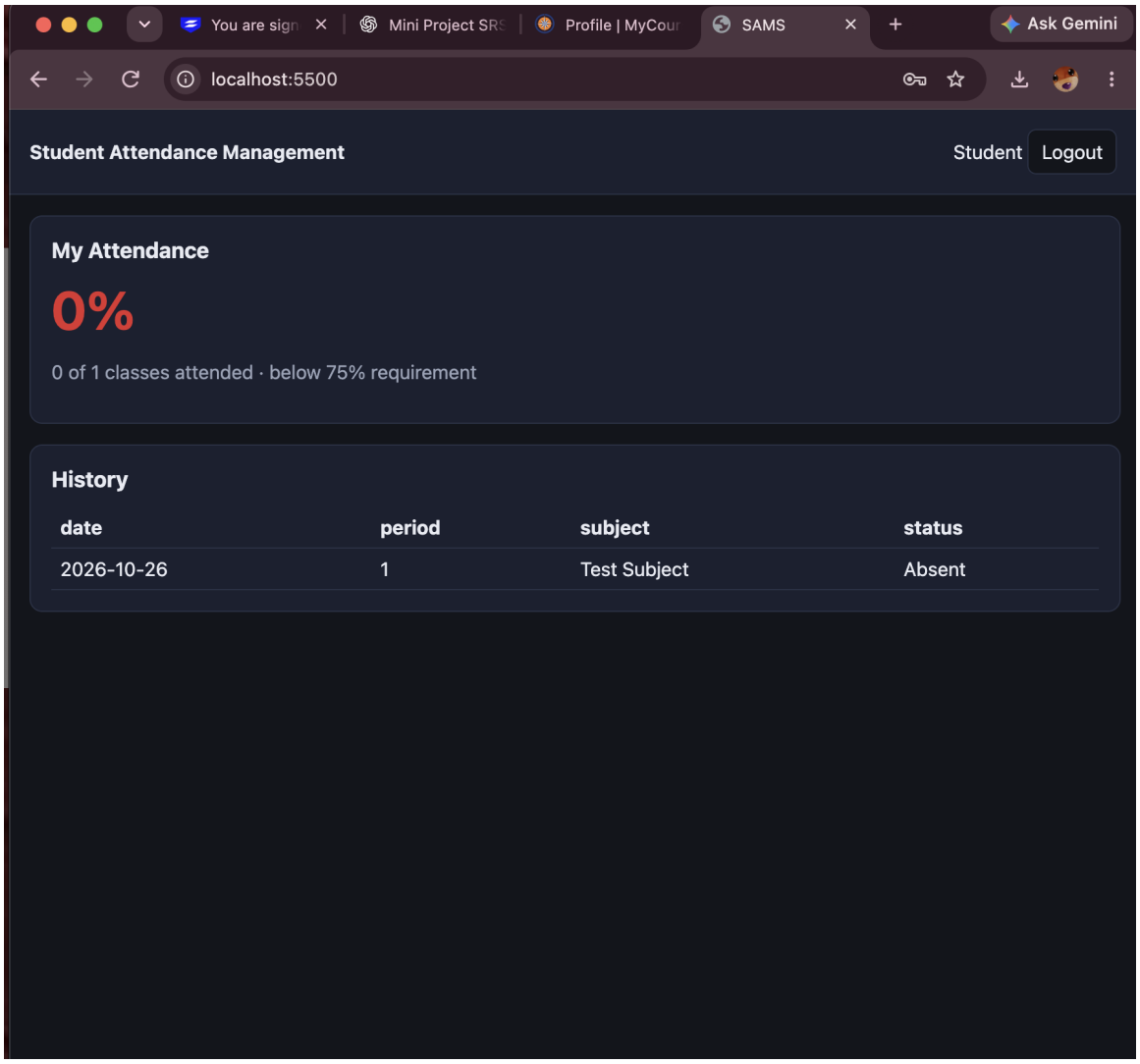
Evidence_40.jpg



```
 -c 'import sys,json; print(json.load(sys.stdin)["token"])') && curl -i "http://
127.0.0.1:8000/api/attendance/report?classId=1&subjectId=1" -H "Authorization: B
earer $TOKEN"
HTTP/1.1 200 OK
date: Mon, 26 Oct 2026 08:12:44 GMT
server: uvicorn
content-length: 144
content-type: application/json

[{"student_id":1,"student":"Test Student One","class":"Test Class A","subject":"
Test Subject","date":"2026-10-26","period":1,"status":"Absent"}]%
```

EV-041 — Student dashboard after valid login



6. Overall conclusion

The local test session demonstrated core attendance workflows, authentication and role restrictions, attendance marking/correction, reports, audit recording, input validation, password hashing and session expiry. Outstanding items are deployment HTTPS verification, separate mobile responsiveness confirmation and a comprehensive log-redaction review. Performance and concurrency results are smoke tests, not formal load tests.

Prepared from recorded test observations. Verify test-to-screenshot mapping and attach/retain original evidence before formal submission.